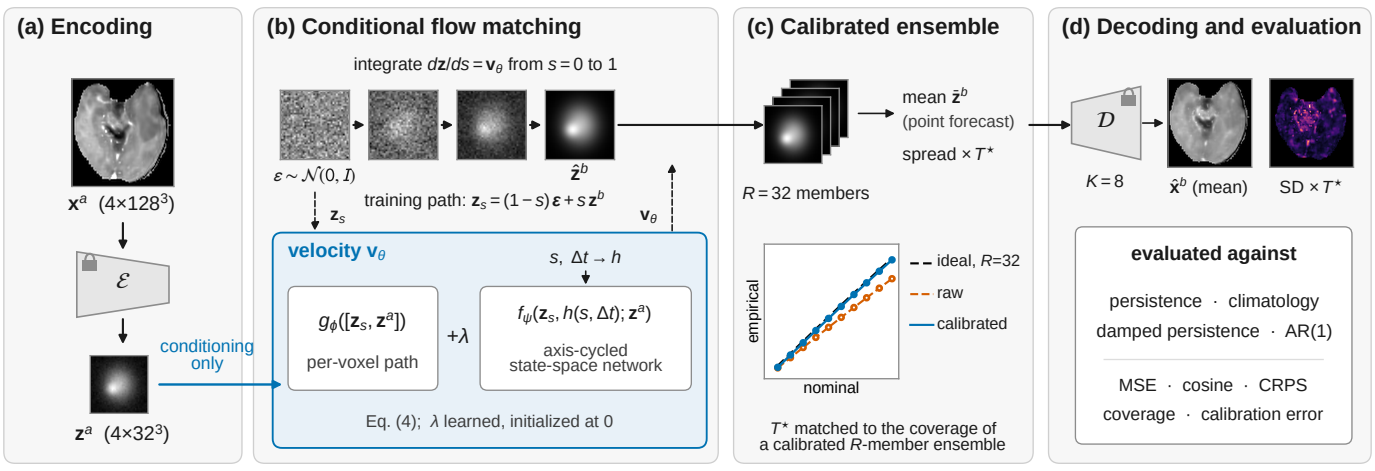

(a) Encoding
$\mathbf{x}^a$ (4×128³)
$\mathcal{E}$
$\mathbf{z}^a$ (4×32³)
conditioning only
(b) Conditional flow matching
integrate $d\mathbf{z}/ds = \mathbf{v}_\theta$ from $s = 0$ to 1
$\varepsilon \sim \mathcal{N}(0, I)$
$\hat{\mathbf{z}}^b$
$\mathbf{z}_s$
training path: $\mathbf{z}_s = (1-s)\,\boldsymbol{\varepsilon} + s\,\mathbf{z}^b$
$\mathbf{v}_\theta$
velocity $\mathbf{v}_\theta$
$s,\ \Delta t \to h$
$g_\phi([\mathbf{z}_s, \mathbf{z}^a])$
per-voxel path
+λ
$f_\psi(\mathbf{z}_s, h(s, \Delta t); \mathbf{z}^a)$
axis-cycled state-space network
Eq. (4); λ learned, initialized at 0
(c) Calibrated ensemble
mean $\bar{\mathbf{z}}^b$
(point forecast)
spread × $T^\star$
$R = 32$ members
empirical
nominal
ideal, R=32
raw
calibrated
$T^\star$ matched to the coverage of a calibrated R-member ensemble
(d) Decoding and evaluation
$\mathcal{D}$
$K = 8$
$\hat{\mathbf{x}}^b$ (mean)
SD × $T^\star$
evaluated against
persistence · climatology
damped persistence · AR(1)
MSE · cosine · CRPS
coverage · calibration error